ComplianceHub™

What is ComplianceHub?

Visibility + Accountability + Action.

ComplianceHub is:

- A platform that helps you manage compliance work effectively & efficiently
- A place where all compliance data is aligned and visible (Single trusted source)
- Built to grow with your operation, system by system
- Proudly South African, locally developed and supported

ComplianceHub Development Roadmap (2025 & 2026)

1. Our Developed Systems [* with continuous expansion]

- Document Management (DMS)
- Document Development (DDS)
- Risk Management (RMS)
- Training Management (TMS) *
- Certificate Management (EPAMS)
- Smart Device (SDS) *

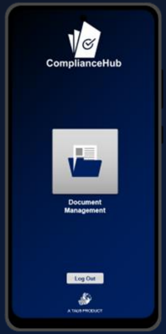
2. Our 2026 System Developments [Progressive L-R]

- Compliance Tracking (CTS)
- Field Inspection (FIS)
- Emergency Response (ERS)
- Asset Management (AMS)
- Control Room Management (CRMS)
- Meeting Management (MMS)

3. Client Driven Developments [Business Case Driven – Affects 2026 Q2+ Timelines]

- Missing Person Response
- Productivity Optimisation (Coal UG)
- Asset Digital Twins
- Full Contractor Management
- All Permits & Certificates

1. Our Developed Systems Full 2025 + Q1 2026



Document Management

Securely stores and manages documents in a centralized repository.

[Storage, Version Controlled, Update Management]

Document Development

Standardized development and updates of documents in client templates.

[Procedures, Standards, SIs]

Risk Management

Provides robust tools to assess, mitigate, monitor and manage operational risks.

[BLRA, IBRA, JRA, Controls]

Training Management

Facilitates the development, tracking, and completion of online training modules.

[Induction, Online, Site Passports]

EPA Management

Securely stores and manages legal compliance certification and digital warehouses.

[Flameproof & Other Certificates]

2. The 2026 Development Overview

Compliance Tracking (CTS)

- Automated and manual task allocation
- Team performance tracking and reporting
- Future: Asset performance & insights generation

Asset Management (AMS)

- Digital twins for TMM and conveyors
- Integration with SAP & other
- Future: Expansion to all asset classes

Field Inspection (FIS)

- Configurable inspection forms across use cases
- Use cases – Inspections, Works Orders, Pre-Use
- Future: Insights from AMS, SAP etc during inspections

Meeting Management (MMS)

- Structured meeting minutes and actions
- Direct linkage to CTS tasks
- Future: Full workflow and system integration

Control Room Management (CRMS)

- Downtime recording and event tracking
- Future: Integration with core operational systems
- Future: Data-driven control integration

Emergency Response (ERS)

- Guided digital emergency response workflows
- Real-time decision support in incidents
- Future: Integrated analytics (MPLS as example)

How Are Systems Linked?

Very Important: Each System can fully function on its own or be linked with any number of other systems.

Document Development

Risk Management

Training Management

Certificate Management

Document Management

Asset Management

Smart Device

Compliance Tracking

Field Inspection

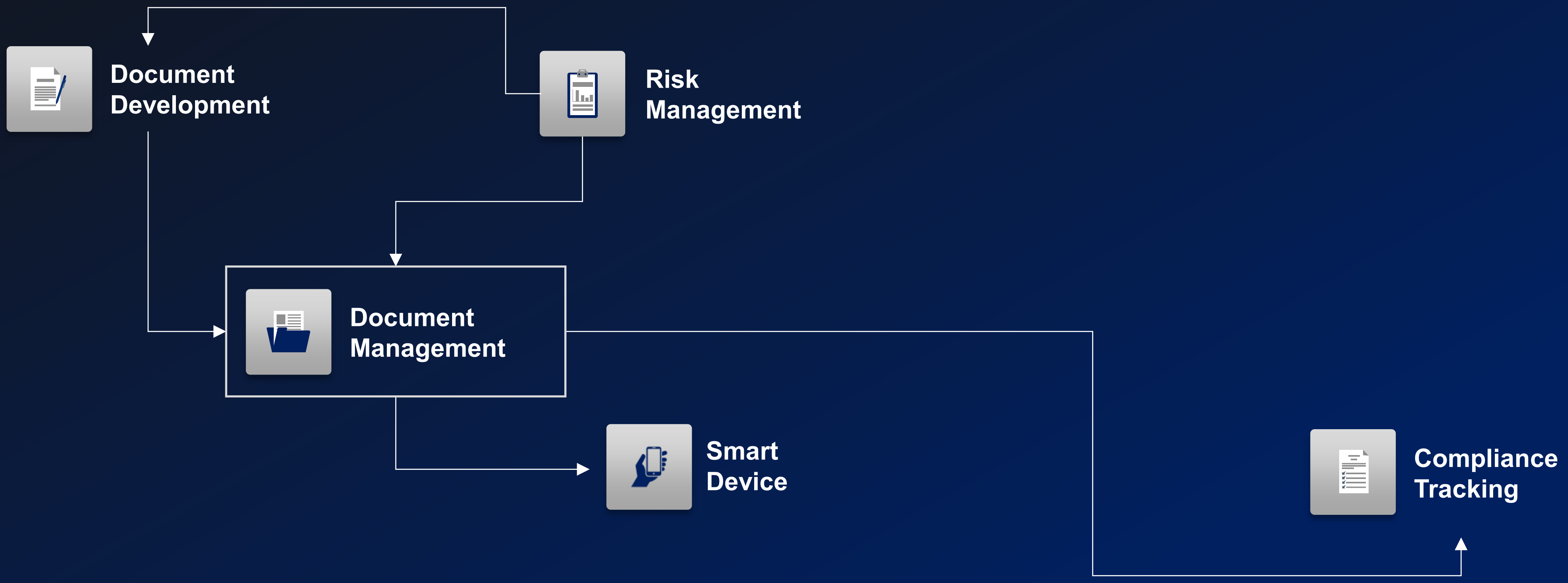
Emergency Response

Control Room Management

Meeting Management

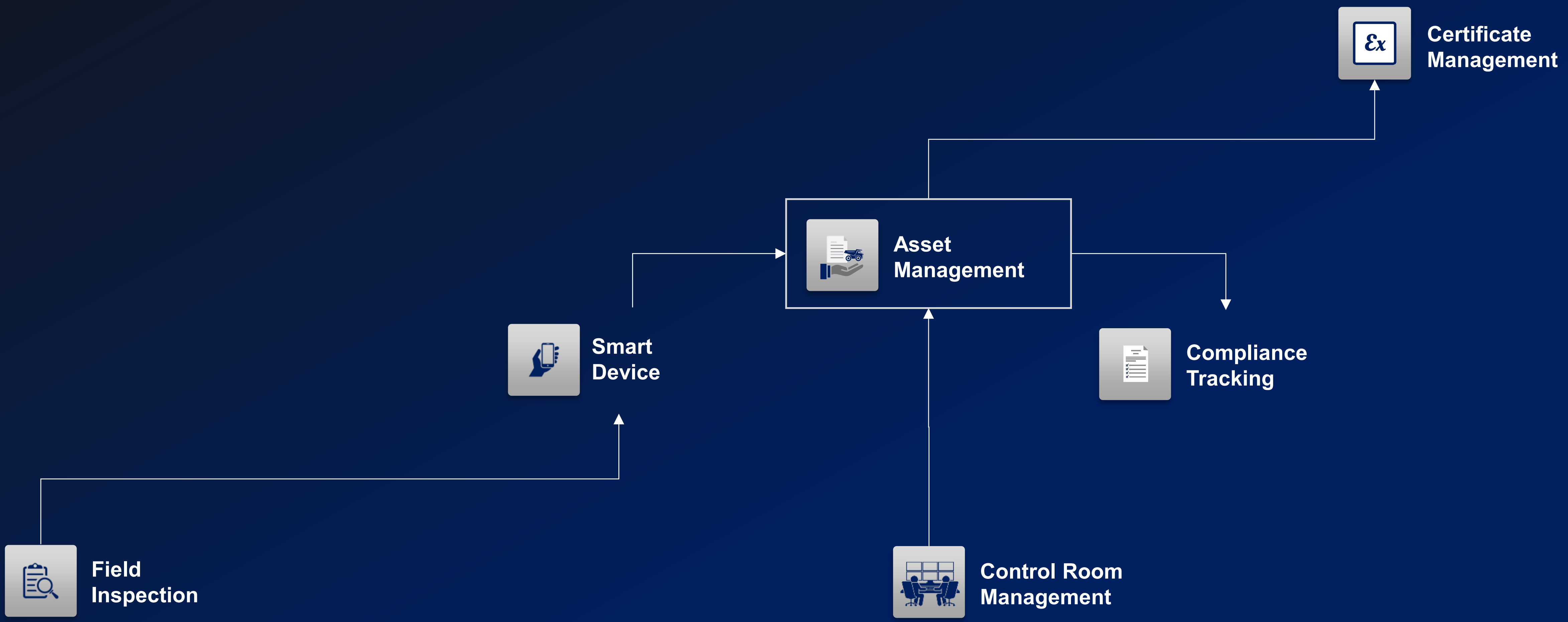
How Are Systems Linked?

Document Management Links



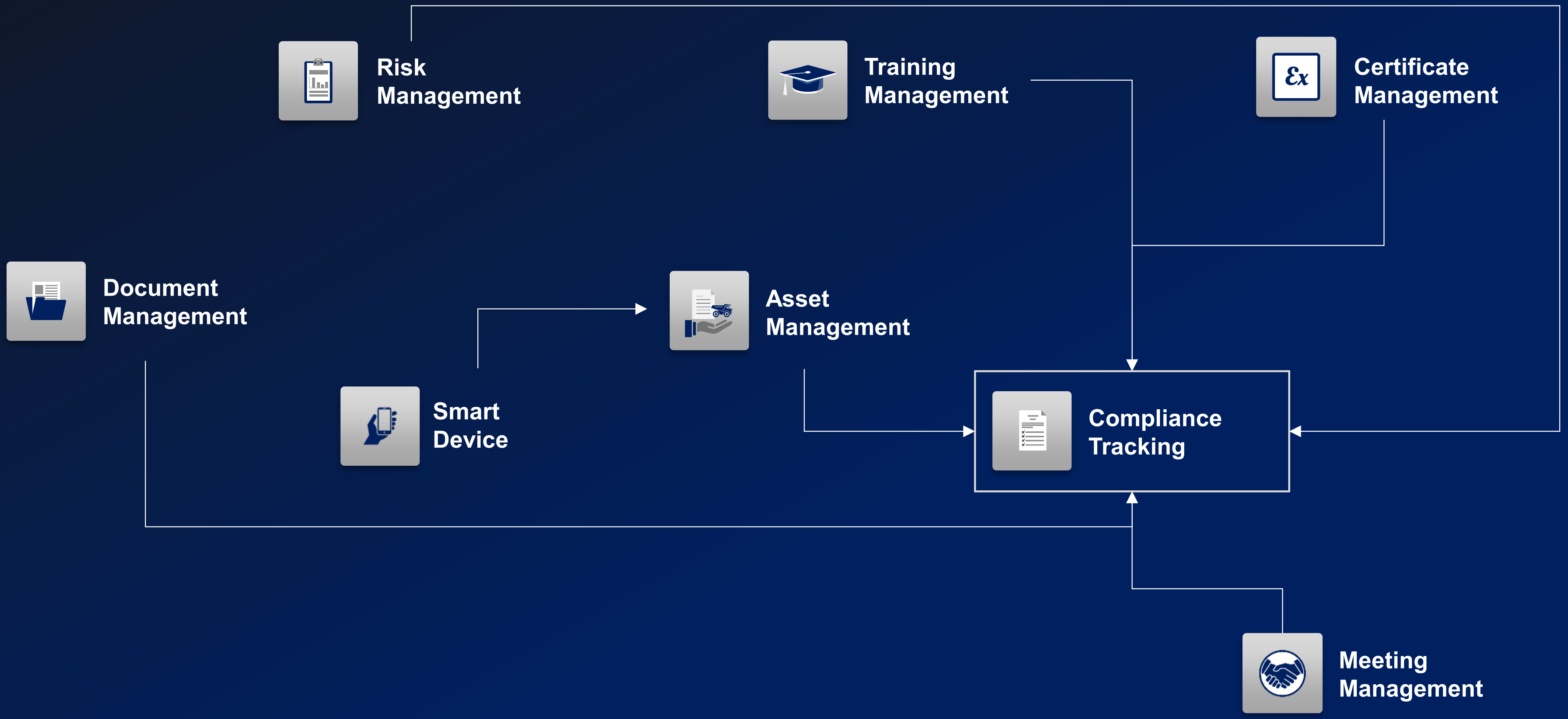
How Are Systems Linked?

Asset Management Links

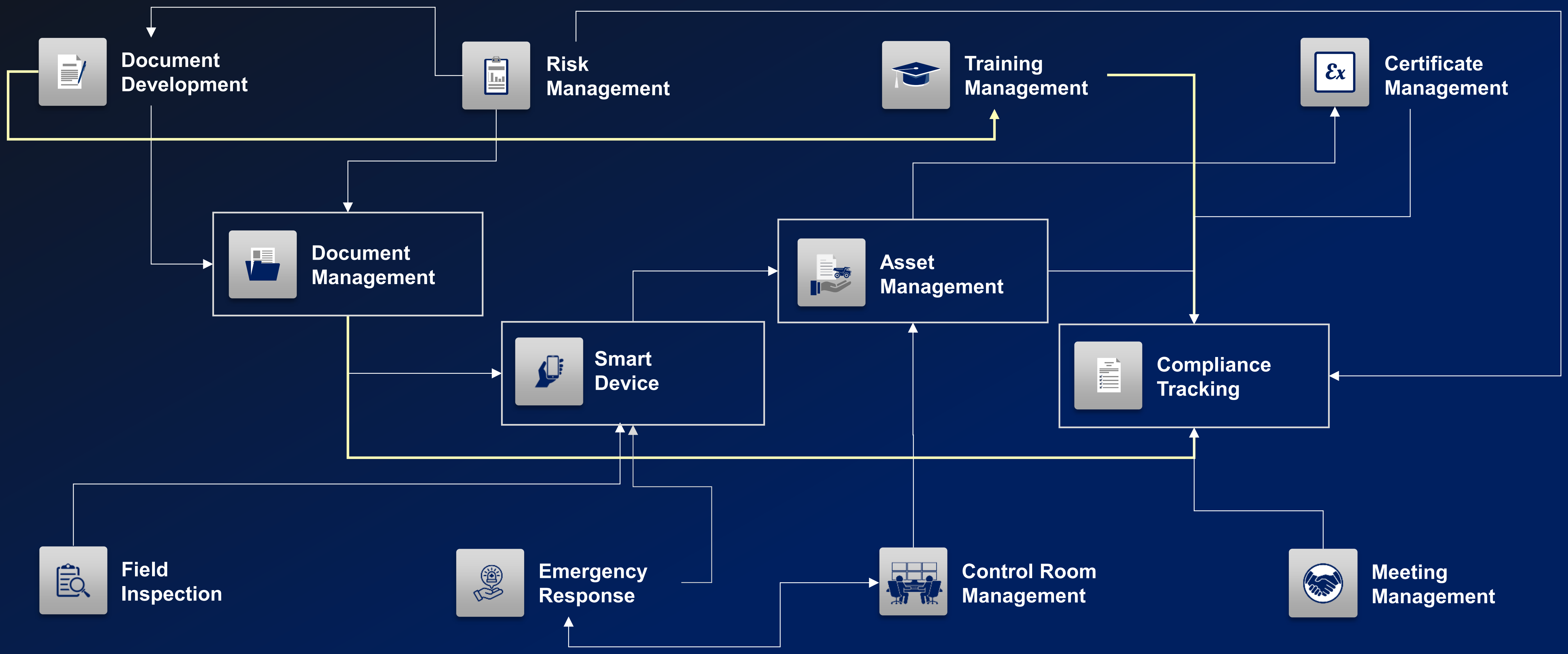


How Are Systems Linked?

Compliance Tracking Links

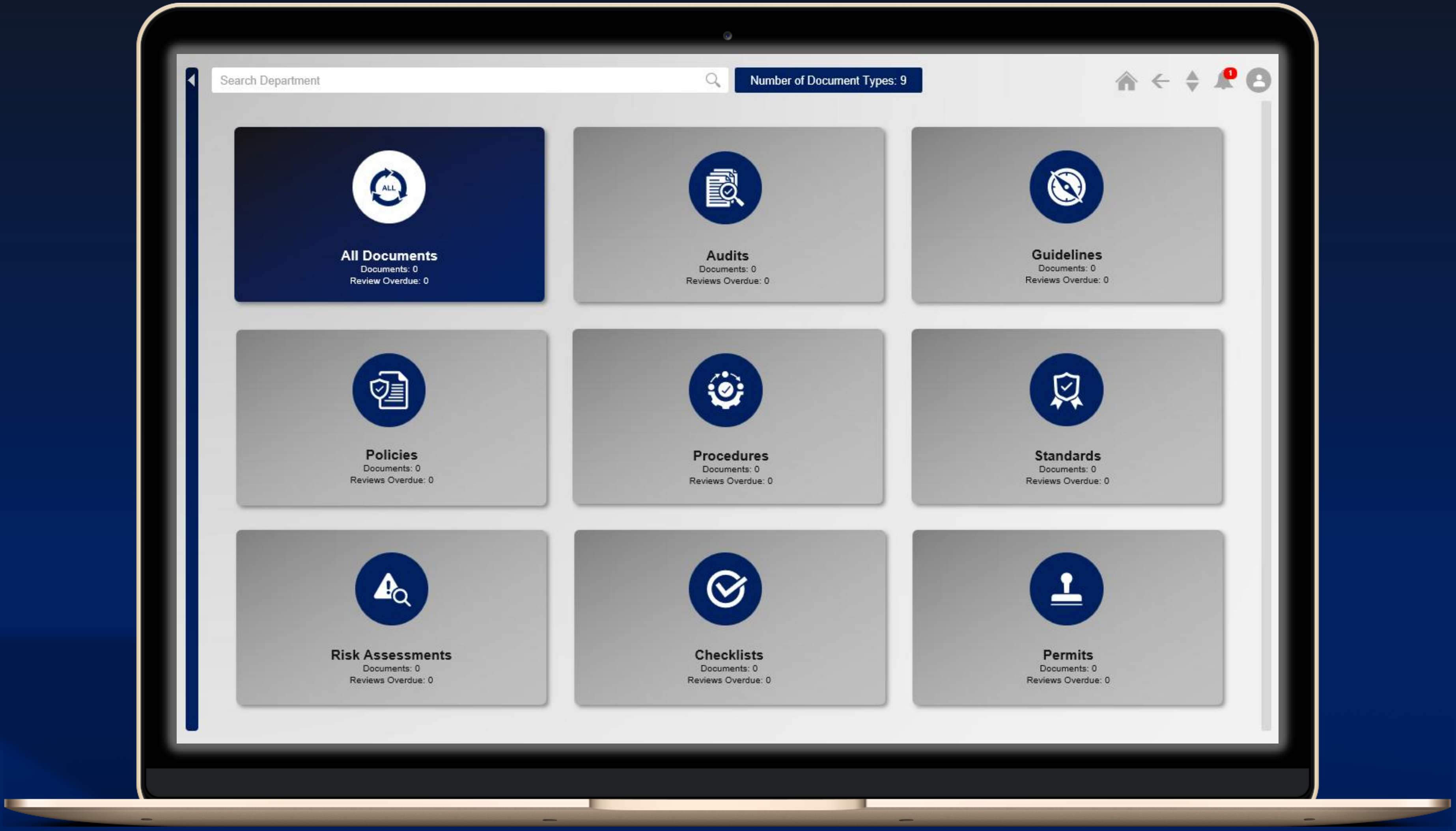


How Are Systems Linked?

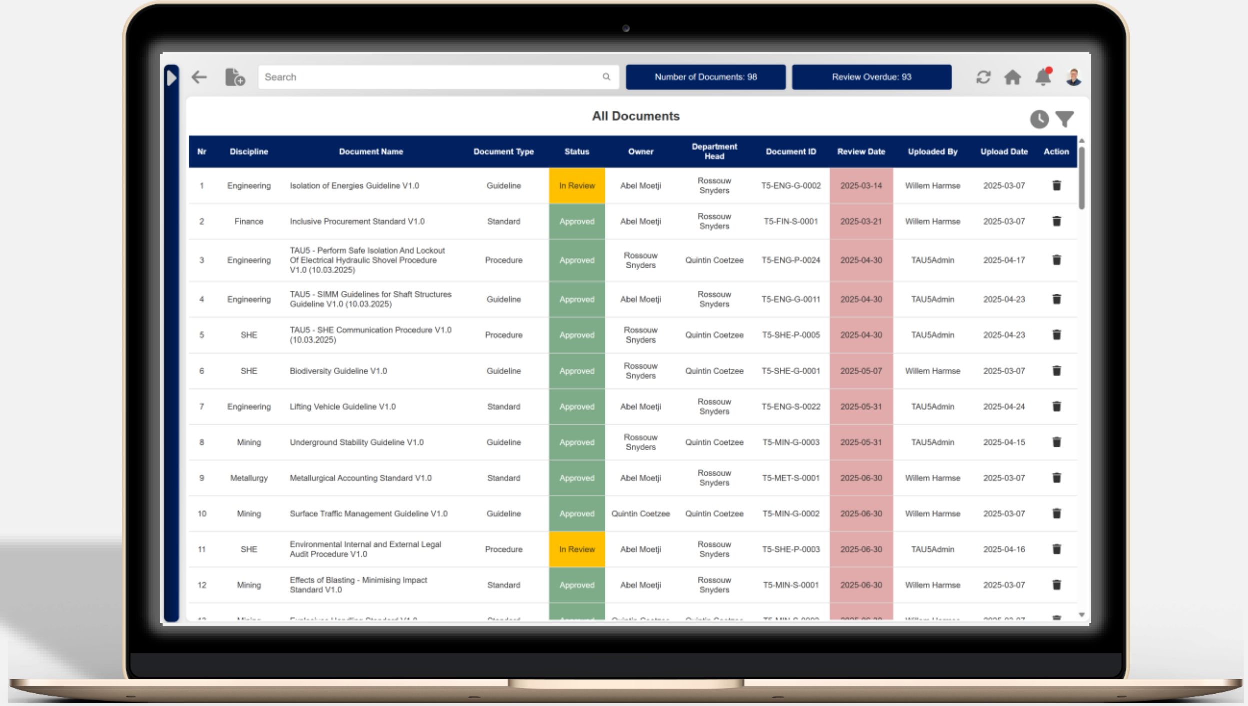


Let’s look at some more detail on developed systems.

Document Management System (DMS)



Document Management System Description



Nr	Discipline	Document Name	Document Type	Status	Owner	Department Head	Document ID	Review Date	Uploaded By	Upload Date	Action
1	Engineering	Isolation of Energies Guideline V1.0	Guideline	In Review	Abel Moetji	Rossouw Snyders	T5-ENG-G-0002	2025-03-14	Willem Harmse	2025-03-07	
2	Finance	Inclusive Procurement Standard V1.0	Standard	Approved	Abel Moetji	Rossouw Snyders	T5-FIN-S-0001	2025-03-21	Willem Harmse	2025-03-07	
3	Engineering	TAU5 - Perform Safe Isolation And Lockout Of Electrical Hydraulic Shovel Procedure V1.0 (10.03.2025)	Procedure	Approved	Rossouw Snyders	Quintin Coetzee	T5-ENG-P-0024	2025-04-30	TAU5Admin	2025-04-17	
4	Engineering	TAU5 - SIMM Guidelines for Shaft Structures Guideline V1.0 (10.03.2025)	Guideline	Approved	Abel Moetji	Rossouw Snyders	T5-ENG-G-0011	2025-04-30	TAU5Admin	2025-04-23	
5	SHE	TAU5 - SHE Communication Procedure V1.0 (10.03.2025)	Procedure	Approved	Rossouw Snyders	Quintin Coetzee	T5-SHE-P-0005	2025-04-30	TAU5Admin	2025-04-23	
6	SHE	Biodiversity Guideline V1.0	Guideline	Approved	Rossouw Snyders	Quintin Coetzee	T5-SHE-G-0001	2025-05-07	Willem Harmse	2025-03-07	
7	Engineering	Lifting Vehicle Guideline V1.0	Standard	Approved	Abel Moetji	Rossouw Snyders	T5-ENG-S-0022	2025-05-31	TAU5Admin	2025-04-24	
8	Mining	Underground Stability Guideline V1.0	Guideline	Approved	Rossouw Snyders	Quintin Coetzee	T5-MIN-G-0003	2025-05-31	TAU5Admin	2025-04-15	
9	Metallurgy	Metallurgical Accounting Standard V1.0	Standard	Approved	Abel Moetji	Rossouw Snyders	T5-MET-S-0001	2025-06-30	Willem Harmse	2025-03-07	
10	Mining	Surface Traffic Management Guideline V1.0	Guideline	Approved	Quintin Coetzee	Quintin Coetzee	T5-MIN-G-0002	2025-06-30	Willem Harmse	2025-03-07	
11	SHE	Environmental Internal and External Legal Audit Procedure V1.0	Procedure	In Review	Abel Moetji	Rossouw Snyders	T5-SHE-P-0003	2025-06-30	TAU5Admin	2025-04-16	
12	Mining	Effects of Blasting - Minimising Impact Standard V1.0	Standard	Approved	Abel Moetji	Rossouw Snyders	T5-MIN-S-0001	2025-06-30	Willem Harmse	2025-03-07	

Centralized Storage:
All documents are stored in a single, structured system, reducing duplication and loss.

Version Management:
Automatically tracks document versions to ensure only the latest approved versions are accessed.

Accessibility:
Allows users to quickly locate documents using keywords, categories, or historical usage.

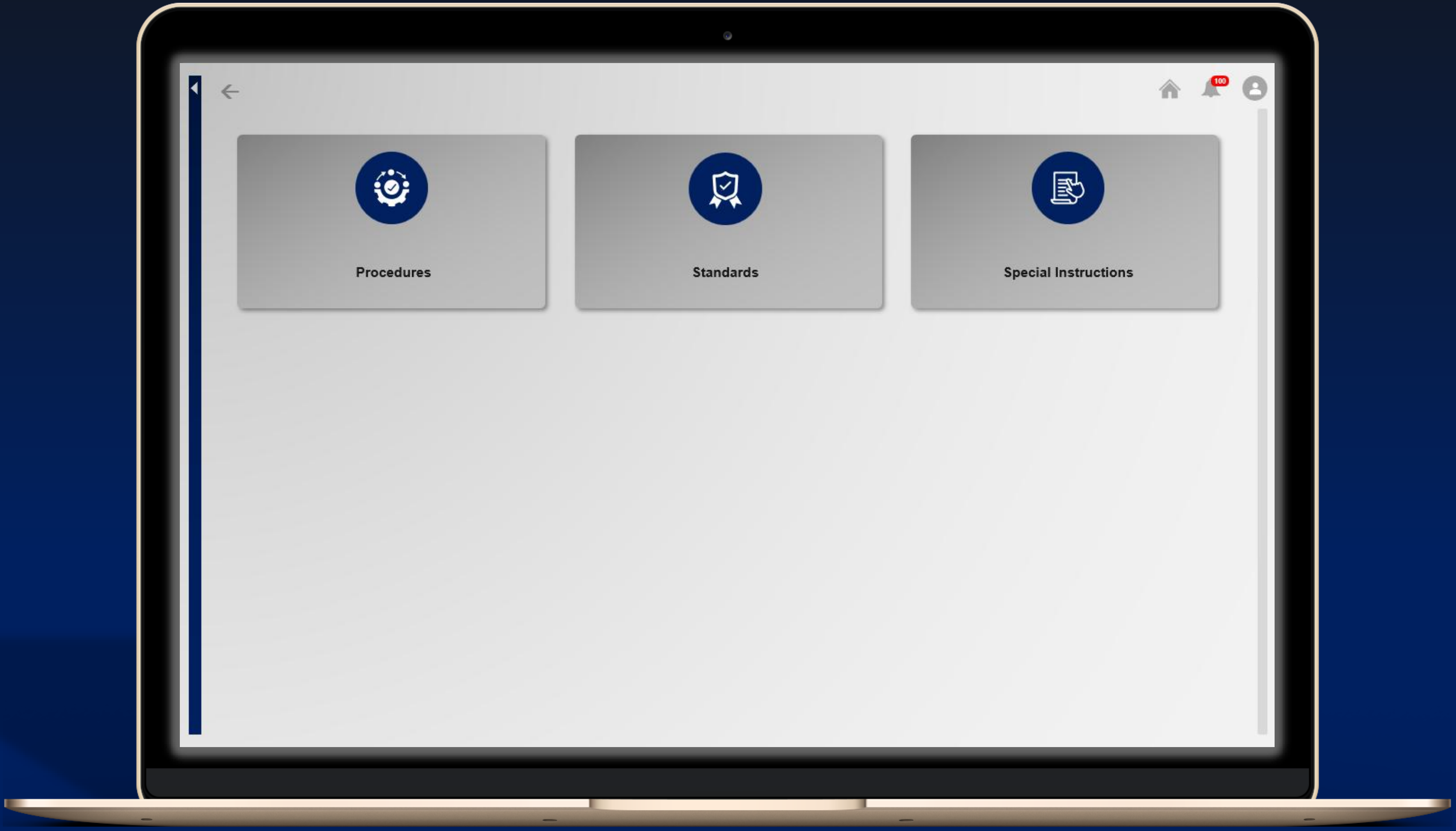
Review & Approval Mechanisms:
Built-in workflows route documents through review and approval steps in an efficient, structured manner.

Digital Sign Off:
Digital signatures and automated notifications simplify approvals.
The functionality to incorporate automated sign off is still in development.

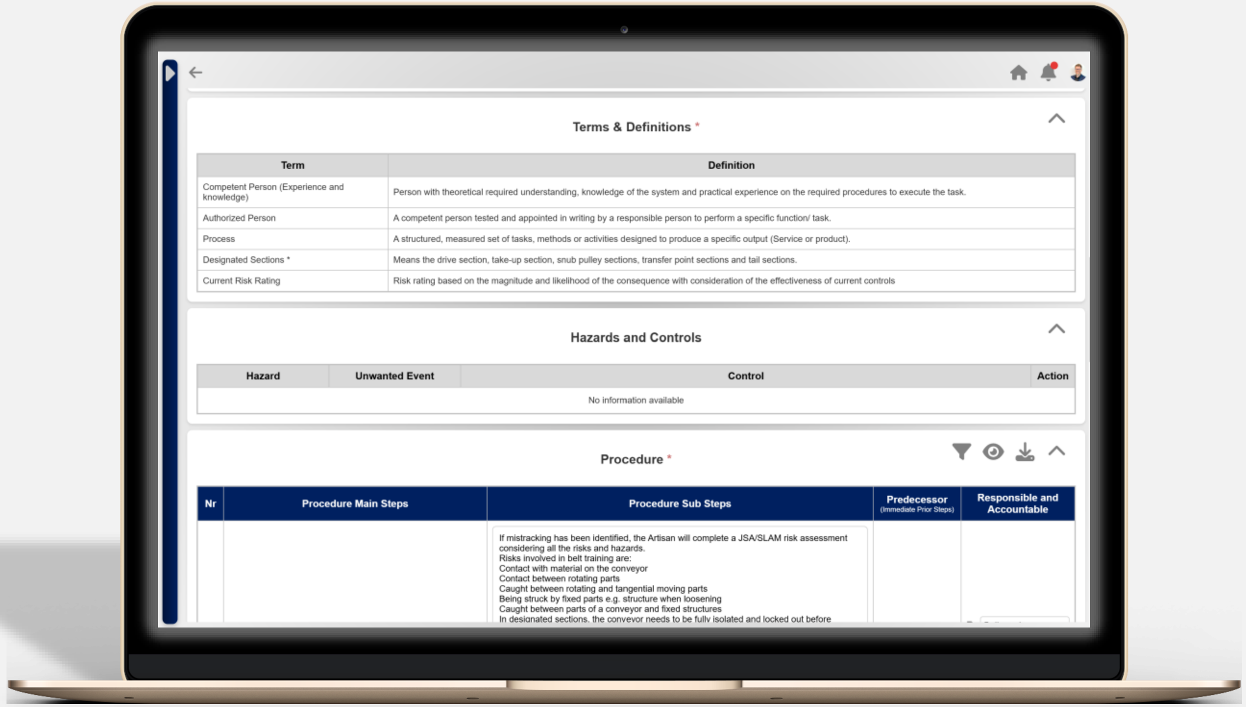
Document Status Management:
Flags documents due for review or removal to maintain compliance.

Mobile Access:
A dedicated Android app enables field staff to access essential documents while on-site, ensuring that critical operational procedures and forms are always within reach.

Document Development System (DDS)



Document Development System Description



Seamless Document Development & Updates:
Enables fast development and updates of professional documents using client specific templates.

Streamlined Editing and Collaboration:
Approved users can collaboratively add content and make live edits, reducing development cycle time.

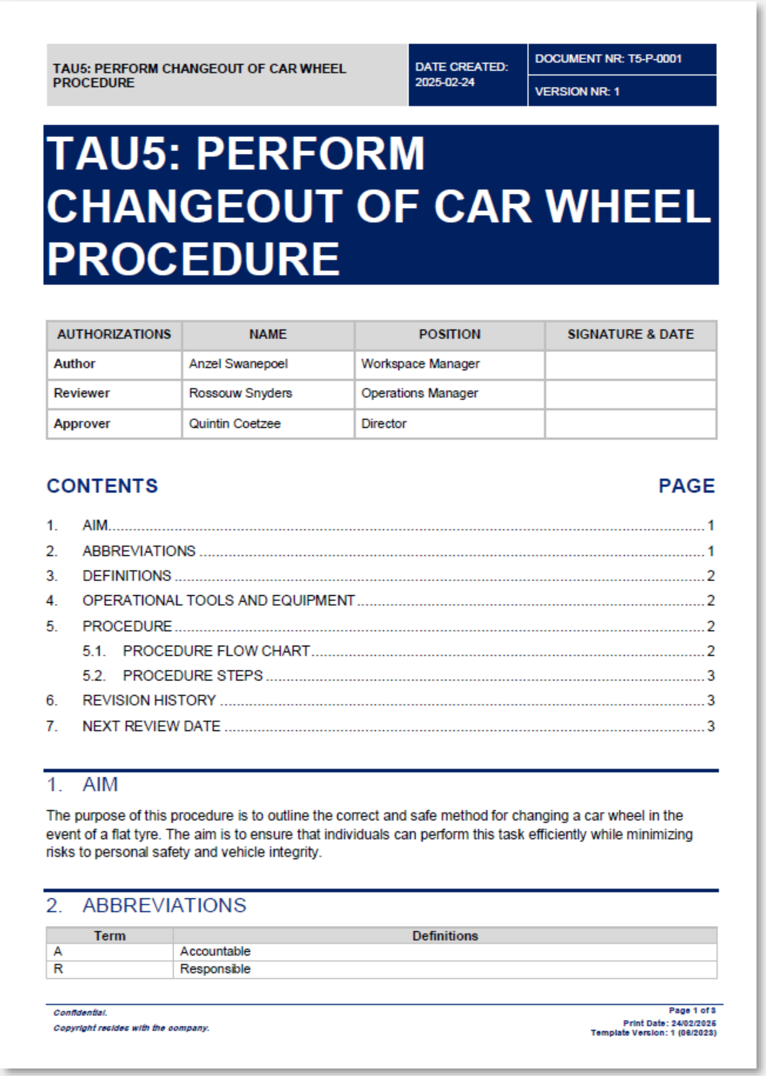
Automated Flow Charts for Procedures:
For documents classified as procedures, the system automatically generates flow charts to visually represent each step.

Pre-approved Standard Information:
Standard elements such as PPE requirements, tools, abbreviations, and definitions are stored in a shared library, minimizing duplication and promoting consistency.

Client-Specific Output Format:
Finalized documents are published directly into the client's Word templates, preserving their layout, branding, and formatting standards.

AI-Powered Language Assistance (Optional):
Integrated with integrated with OpenAI's GPR4.O to allow users to rewrite or refine sections of documents, improving clarity, grammar, and tone where needed, only if the client enables this functionality.

Example: DDS Procedure Output



TAU5: PERFORM CHANGEOUT OF CAR WHEEL PROCEDURE	DATE CREATED: 2025-02-24	DOCUMENT NR: T5-P-0001
		VERSION NR: 1

TAU5: PERFORM CHANGEOUT OF CAR WHEEL PROCEDURE

AUTHORIZATIONS	NAME	POSITION	SIGNATURE & DATE
Author	Anzel Swanepoel	Workspace Manager	
Reviewer	Rossouw Snyders	Operations Manager	
Approver	Quintin Coetzee	Director	

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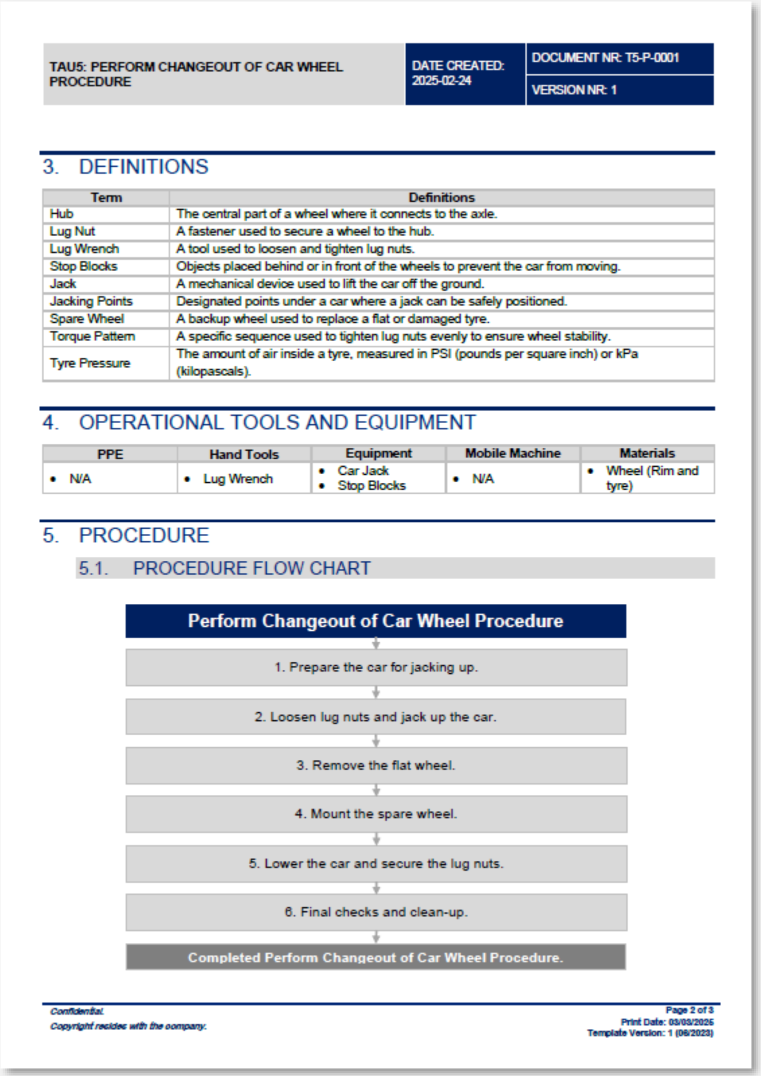
1. AIM

The purpose of this procedure is to outline the correct and safe method for changing a car wheel in the event of a flat tyre. The aim is to ensure that individuals can perform this task efficiently while minimizing risks to personal safety and vehicle integrity.

2. ABBREVIATIONS

Term	Definitions
A	Accountable
R	Responsible



TAU5: PERFORM CHANGEOUT OF CAR WHEEL PROCEDURE	DATE CREATED: 2025-02-24	DOCUMENT NR: T5-P-0001
		VERSION NR: 1

3. DEFINITIONS

Term	Definitions
Hub	The central part of a wheel where it connects to the axle.
Lug Nut	A fastener used to secure a wheel to the hub.
Lug Wrench	A tool used to loosen and tighten lug nuts.
Stop Blocks	Objects placed behind or in front of the wheels to prevent the car from moving.
Jack	A mechanical device used to lift the car off the ground.
Jacking Points	Designated points under a car where a jack can be safely positioned.
Spare Wheel	A backup wheel used to replace a flat or damaged tyre.
Torque Pattern	A specific sequence used to tighten lug nuts evenly to ensure wheel stability.
Tyre Pressure	The amount of air inside a tyre, measured in PSI (pounds per square inch) or kPa (kilopascals).

4. OPERATIONAL TOOLS AND EQUIPMENT

PPE	Hand Tools	Equipment	Mobile Machine	Materials
• N/A	• Lug Wrench	• Car Jack • Stop Blocks	• N/A	• Wheel (Rim and tyre)

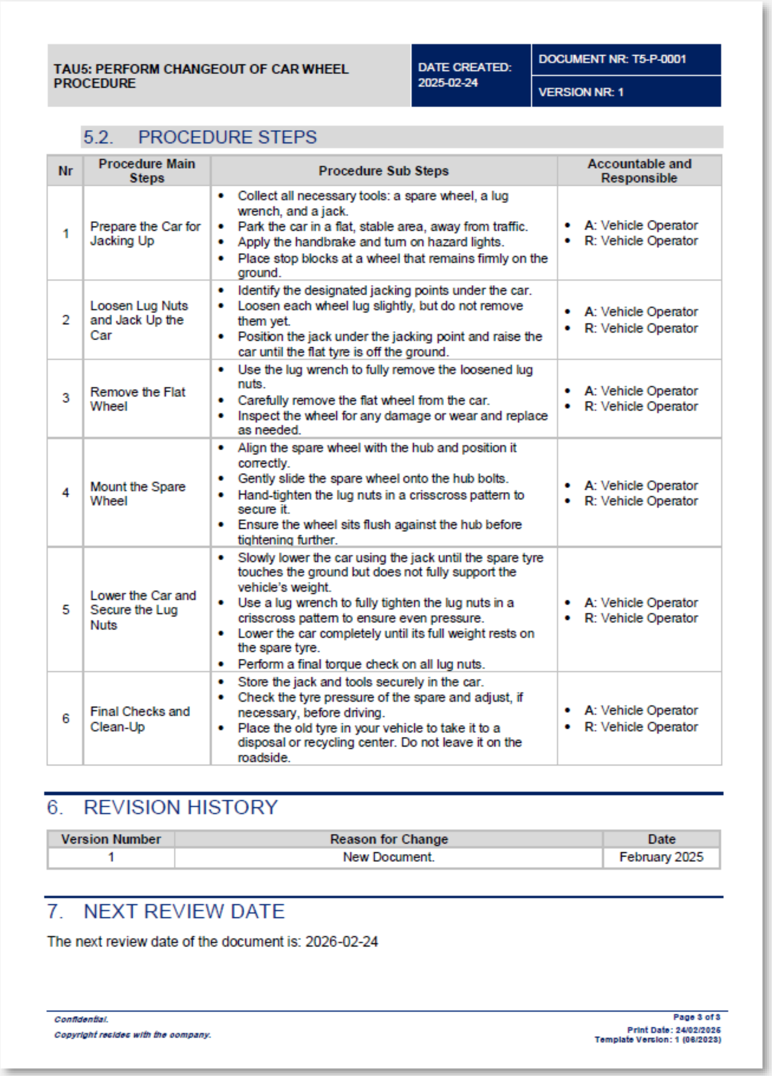
5. PROCEDURE

5.1. PROCEDURE FLOW CHART

Perform Changeout of Car Wheel Procedure

1. Prepare the car for jacking up.
2. Loosen lug nuts and jack up the car.
3. Remove the flat wheel.
4. Mount the spare wheel.
5. Lower the car and secure the lug nuts.
6. Final checks and clean-up.

Completed Perform Changeout of Car Wheel Procedure.



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		VERSION NR: 1

5.2. PROCEDURE STEPS

Nr	Procedure Main Steps	Procedure Sub Steps	Accountable and Responsible
1	Prepare the Car for Jacking Up	• Collect all necessary tools: a spare wheel, a lug wrench, and a jack. • Park the car in a flat, stable area, away from traffic. • Apply the handbrake and turn on hazard lights. • Place stop blocks at a wheel that remains firmly on the ground.	• A: Vehicle Operator • R: Vehicle Operator
2	Loosen Lug Nuts and Jack Up the Car	• Identify the designated jacking points under the car. • Loosen each wheel lug slightly, but do not remove them yet. • Position the jack under the jacking point and raise the car until the flat tyre is off the ground.	• A: Vehicle Operator • R: Vehicle Operator
3	Remove the Flat Wheel	• Use the lug wrench to fully remove the loosened lug nuts. • Carefully remove the flat wheel from the car. • Inspect the wheel for any damage or wear and replace as needed.	• A: Vehicle Operator • R: Vehicle Operator
4	Mount the Spare Wheel	• Align the spare wheel with the hub and position it correctly. • Gently slide the spare wheel onto the hub bolts. • Hand-tighten the lug nuts in a crisscross pattern to secure it. • Ensure the wheel sits flush against the hub before tightening further.	• A: Vehicle Operator • R: Vehicle Operator
5	Lower the Car and Secure the Lug Nuts	• Slowly lower the car using the jack until the spare tyre touches the ground but does not fully support the vehicle's weight. • Use a lug wrench to fully tighten the lug nuts in a crisscross pattern to ensure even pressure. • Lower the car completely until its full weight rests on the spare tyre. • Perform a final torque check on all lug nuts.	• A: Vehicle Operator • R: Vehicle Operator
6	Final Checks and Clean-Up	• Store the jack and tools securely in the car. • Check the tyre pressure of the spare and adjust, if necessary, before driving. • Place the old tyre in your vehicle to take it to a disposal or recycling center. Do not leave it on the roadside.	• A: Vehicle Operator • R: Vehicle Operator

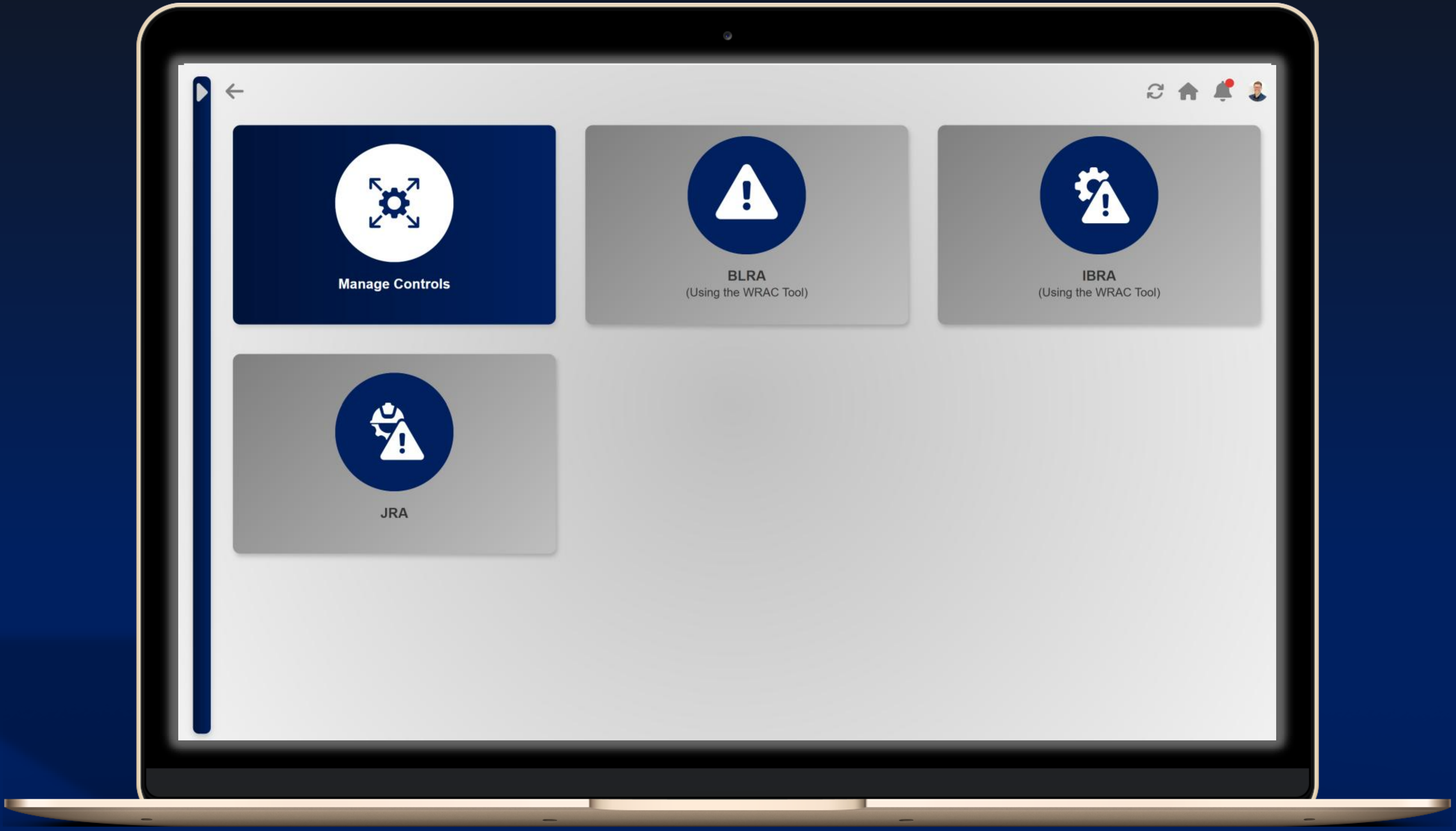
6. REVISION HISTORY

Version Number	Reason for Change	Date
1	New Document.	February 2025

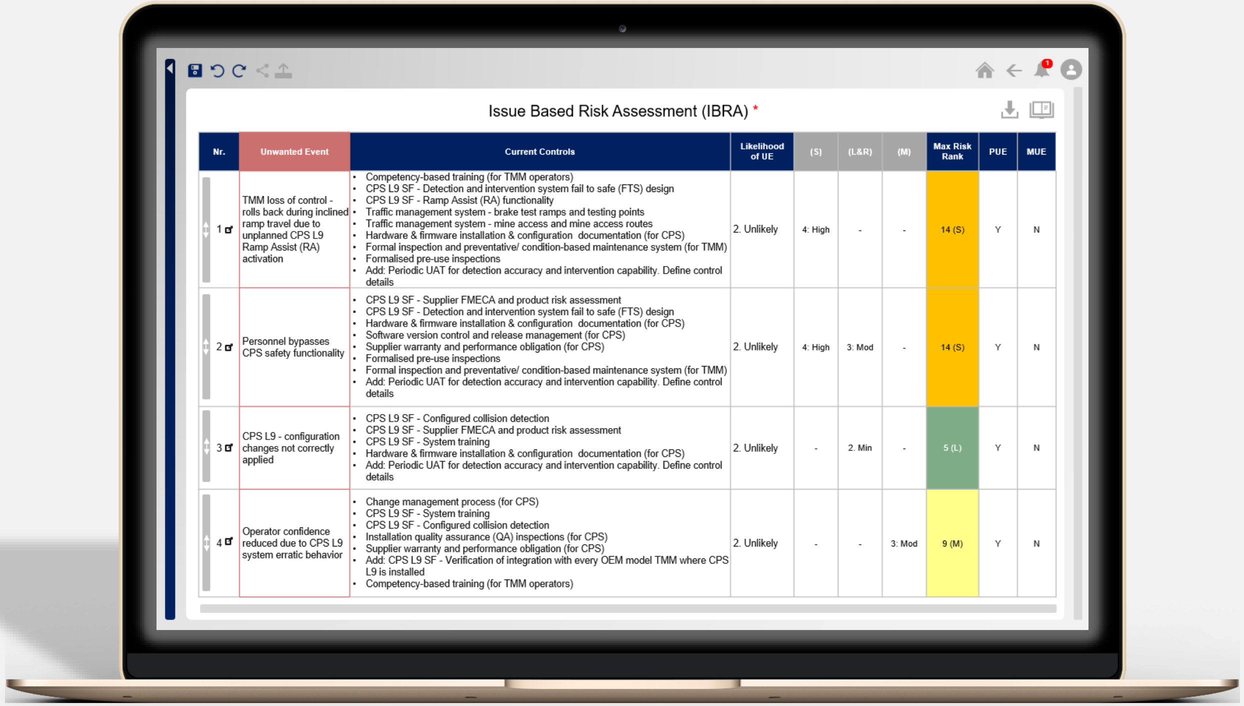
7. NEXT REVIEW DATE

The next review date of the document is: 2026-02-24

Risk Management System (RMS)



Risk Management System Description



Issue Based Risk Assessment (IBRA) *

Nr.	Unwanted Event	Current Controls	Likelihood of UE	(S)	(L&R)	(M)	Max Risk Rank	PUE	MUE
1	TMM loss of control - rolls back during inclined ramp travel due to unplanned CPS L9 Ramp Assist (RA) activation	• Competency-based training (for TMM operators) • CPS L9 SF - Detection and intervention system fail to safe (FTS) design • CPS L9 SF - Ramp Assist (RA) functionality • Traffic management system - brake test ramps and testing points • Traffic management system - mine access and mine access routes • Hardware & firmware installation & configuration documentation (for CPS) • Formal inspection and preventative/ condition-based maintenance system (for TMM) • Formalised pre-use inspections • Add: Periodic UAT for detection accuracy and intervention capability. Define control details	2. Unlikely	4: High	-	-	14 (S)	Y	N
2	Personnel bypasses CPS safety functionality	• CPS L9 SF - Supplier FMECA and product risk assessment • CPS L9 SF - Detection and intervention system fail to safe (FTS) design • Hardware & firmware installation & configuration documentation (for CPS) • Software version control and release management (for CPS) • Supplier warranty and performance obligation (for CPS) • Formalised pre-use inspections • Formal inspection and preventative/ condition-based maintenance system (for TMM) • Add: Periodic UAT for detection accuracy and intervention capability. Define control details	2. Unlikely	4: High	3: Mod	-	14 (S)	Y	N
3	CPS L9 - configuration changes not correctly applied	• CPS L9 SF - Configured collision detection • CPS L9 SF - Supplier FMECA and product risk assessment • CPS L9 SF - System training • Hardware & firmware installation & configuration documentation (for CPS) • Add: Periodic UAT for detection accuracy and intervention capability. Define control details	2. Unlikely	-	2. Min	-	5 (L)	Y	N
4	Operator confidence reduced due to CPS L9 system erratic behavior	• Change management process (for CPS) • CPS L9 SF - System training • CPS L9 SF - Configured collision detection • Installation quality assurance (QA) inspections (for CPS) • Supplier warranty and performance obligation (for CPS) • Add: CPS L9 SF - Verification of integration with every OEM model TMM where CPS L9 is installed • Competency-based training (for TMM operators)	2. Unlikely	-	-	3: Mod	9 (M)	Y	N

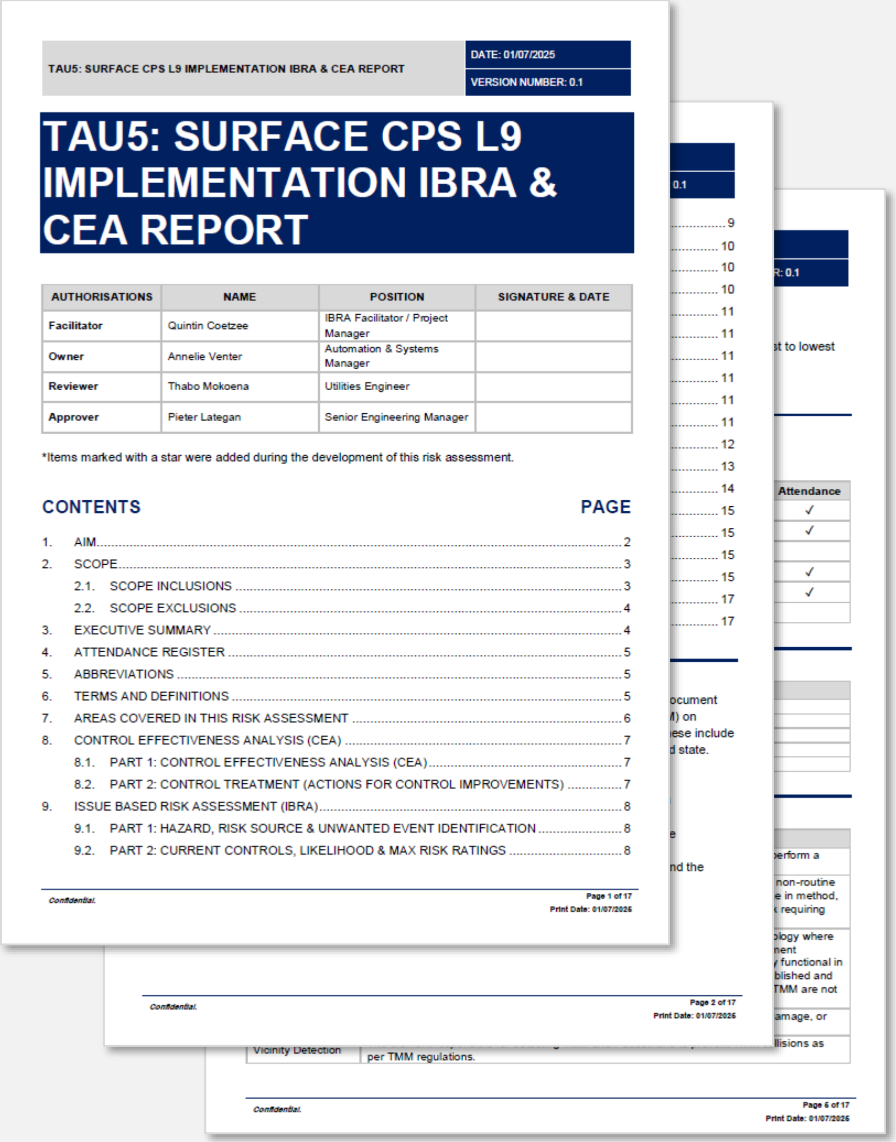
Comprehensive Risk Management Tools:
Provides configurable fields to support the structured identification, evaluation, and mitigation of operational risks.

Standardised Controls:
Approved control measures are listed and standardized across the system, improving consistency and reducing the potential for misuse or ambiguous control descriptions.

Internationally Aligned Risk Assessment Tools:
The system supports various formal formats used across mining and industrial operations: Issue-Based Risk Assessments (IBRA), Baseline Risk Assessments (BLRA), Job Risk Analysis (JRA), Bowtie Risk Assessments (BTA), Control Effectiveness Analysis (CEA).

Standardised Elements for Consistency:
The RMS includes pre-listed controls and standardised unwanted events to enable a more structured and consistent approach to risk documentation. However, users retain flexibility to add new elements, which are flagged for review and approval.

Example: RMS - IBRA Output Report


TAU5: SURFACE CPS L9 IMPLEMENTATION IBRA & CEA REPORT | DATE: 01/07/2025 | VERSION NUMBER: 0.1

TAU5: SURFACE CPS L9 IMPLEMENTATION IBRA & CEA REPORT

AUTHORISATIONS	NAME	POSITION	SIGNATURE & DATE
Facilitator	Quintin Coetzee	IBRA Facilitator / Project Manager	
Owner	Annelie Venter	Automation & Systems Manager	
Reviewer	Thabo Mokoena	Utilities Engineer	
Approver	Pieter Lategan	Senior Engineering Manager	

*Items marked with a star were added during the development of this risk assessment.

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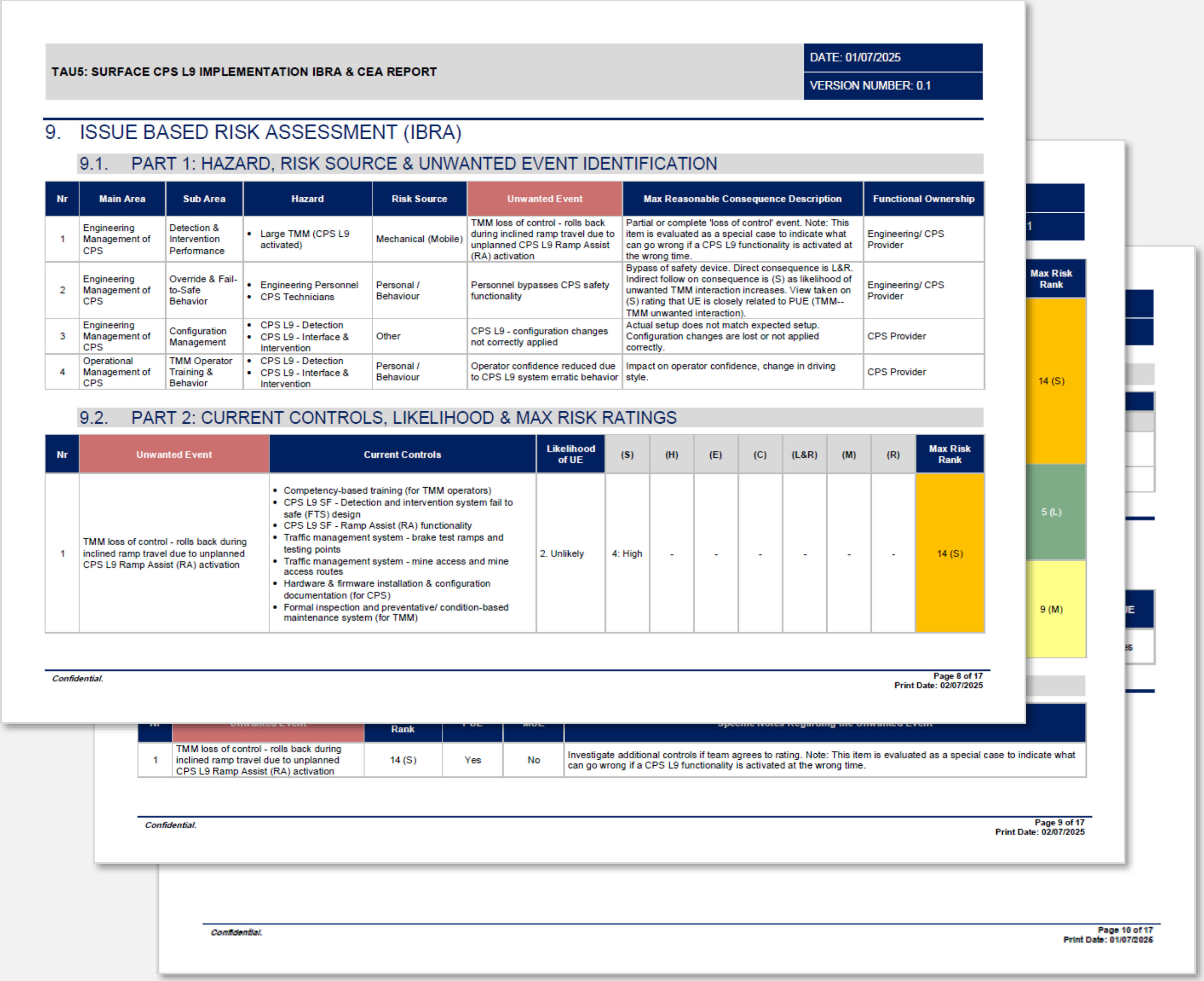

TAU5: SURFACE CPS L9 IMPLEMENTATION IBRA & CEA REPORT | DATE: 01/07/2025 | VERSION NUMBER: 0.1

9. ISSUE BASED RISK ASSESSMENT (IBRA)

9.1. PART 1: HAZARD, RISK SOURCE & UNWANTED EVENT IDENTIFICATION

Nr	Main Area	Sub Area	Hazard	Risk Source	Unwanted Event	Max Reasonable Consequence Description	Functional Ownership
1	Engineering Management of CPS	Detection & Intervention Performance	• Large TMM (CPS L9 activated)	Mechanical (Mobile)	TMM loss of control - rolls back during inclined ramp travel due to unplanned CPS L9 Ramp Assist (RA) activation	Partial or complete 'loss of control' event. Note: This item is evaluated as a special case to indicate what can go wrong if a CPS L9 functionality is activated at the wrong time.	Engineering/ CPS Provider
2	Engineering Management of CPS	Override & Fail-to-Safe Behavior	• Engineering Personnel • CPS Technicians	Personal / Behaviour	Personnel bypasses CPS safety functionality	Bypass of safety device. Direct consequence is L&R. Indirect follow on consequence is (S) as likelihood of unwanted TMM interaction increases. View taken on (S) rating that UE is closely related to PUE (TMM--TMM unwanted interaction).	Engineering/ CPS Provider
3	Engineering Management of CPS	Configuration Management	• CPS L9 - Detection • CPS L9 - Interface & Intervention	Other	CPS L9 - configuration changes not correctly applied	Actual setup does not match expected setup. Configuration changes are lost or not applied correctly.	CPS Provider
4	Operational Management of CPS	TMM Operator Training & Behavior	• CPS L9 - Detection • CPS L9 - Interface & Intervention	Personal / Behaviour	Operator confidence reduced due to CPS L9 system erratic behavior	Impact on operator confidence, change in driving style.	CPS Provider

9.2. PART 2: CURRENT CONTROLS, LIKELIHOOD & MAX RISK RATINGS

Nr	Unwanted Event	Current Controls	Likelihood of UE	(S)	(H)	(E)	(C)	(L&R)	(M)	(R)	Max Risk Rank
1	TMM loss of control - rolls back during inclined ramp travel due to unplanned CPS L9 Ramp Assist (RA) activation	• Competency-based training (for TMM operators) • CPS L9 SF - Detection and intervention system fail to safe (FTS) design • CPS L9 SF - Ramp Assist (RA) functionality • Traffic management system - brake test ramps and testing points • Traffic management system - mine access and mine access routes • Hardware & firmware installation & configuration documentation (for CPS) • Formal inspection and preventative/ condition-based maintenance system (for TMM)	2. Unlikely	4: High	-	-	-	-	-	-	14 (S)

Training Management System (TMS)

Training Management System Description

Custom Training Module Creation:
Clients can create their own training modules.

Rich Content Integration:
Text, images, and video content can be embedded into training modules to enhance engagement and comprehension.

Structured Training Flows:
Training can be sequenced and assigned according to best practice models based on job roles, operational zones, or compliance categories.

Competency Assessments:
Integrated testing and evaluation tools allow clients to assess user understanding through quizzes or formal assessments.

Training Records Management:
Records and stores all user interactions, whether by visitors or employees, including completions and results, ensuring a complete training history is maintained for each user.

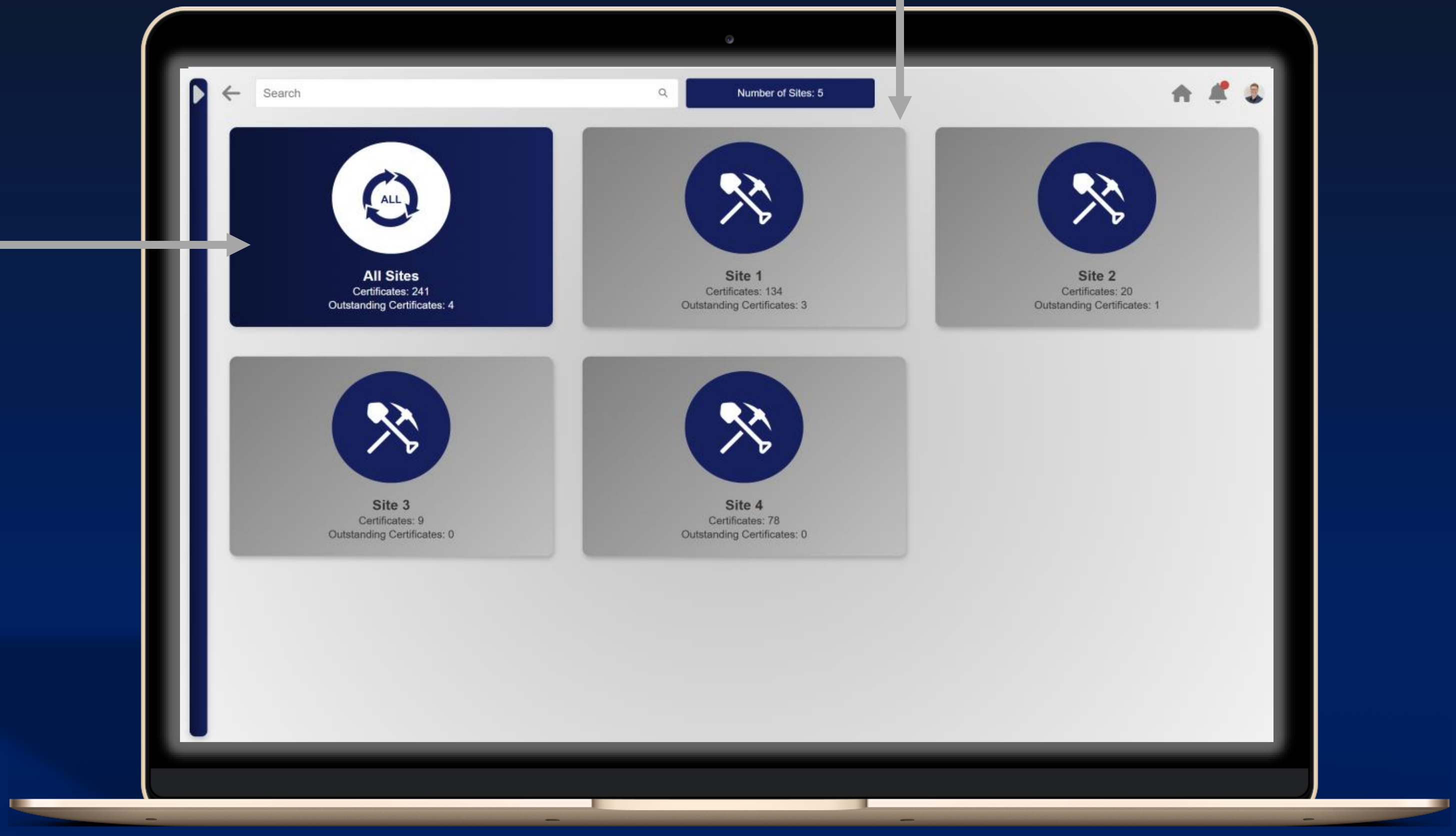
Additional Features:

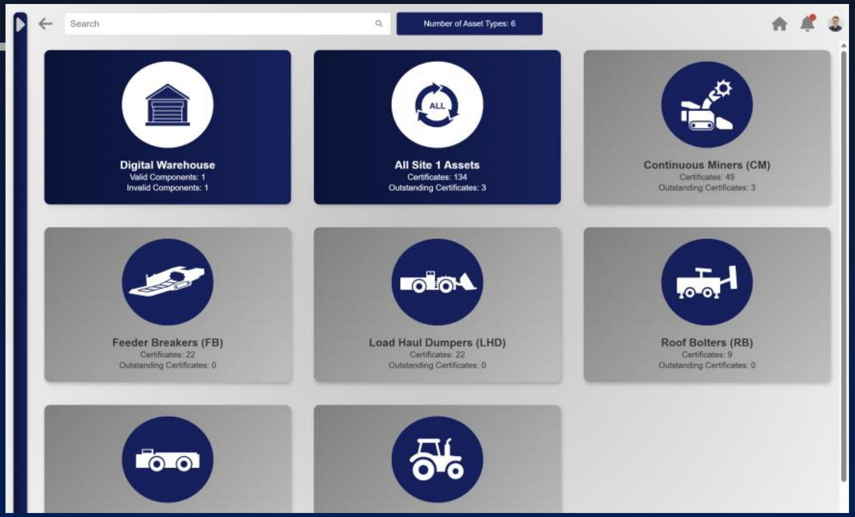
- Auto-reminders for retraining intervals or incomplete modules.
- Certificate generation for completed modules.
- Role-based training paths to support workforce segmentation.

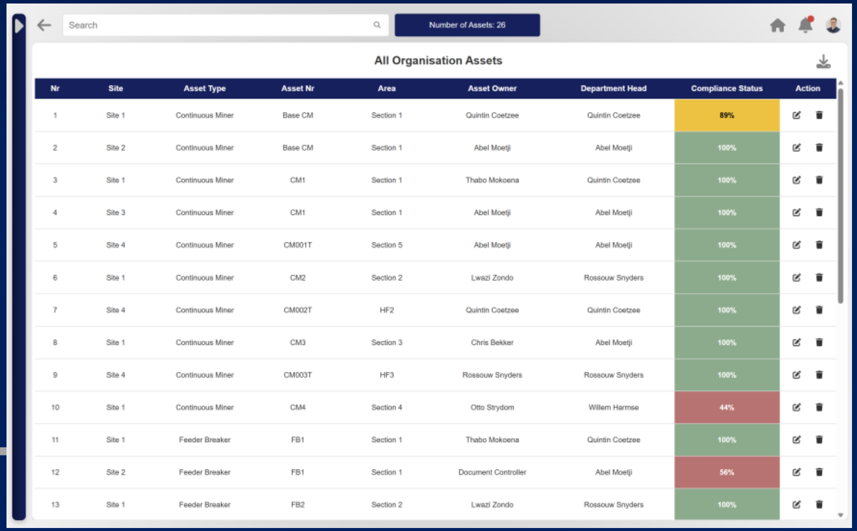
Future Updates:

- Auto-generating content from developed documents.
- Training modules directly linked to the latest versions of relevant procedures and operational documents developed within the system.
- Mobile training access for remote or field-based teams.

Explosion Protected Apparatus Management System (EPAMS)



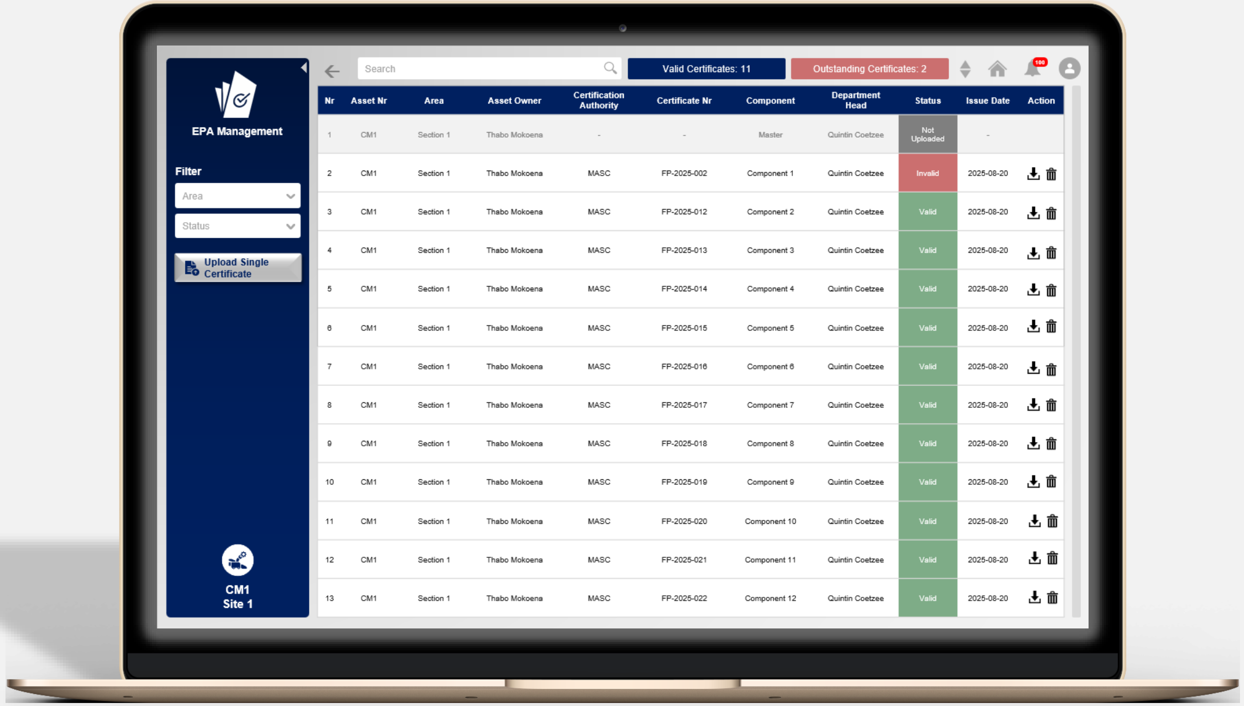




All Organisation Assets

Nr	Site	Asset Type	Asset Nr	Area	Asset Owner	Department Head	Compliance Status	Action
1	Site 1	Continuous Miner	Base CM	Section 1	Quintin Coetzee	Quintin Coetzee	89%	
2	Site 2	Continuous Miner	Base CM	Section 1	Abel Moetji	Abel Moetji	100%	
3	Site 1	Continuous Miner	CM1	Section 1	Thabo Mokoena	Quintin Coetzee	100%	
4	Site 3	Continuous Miner	CM1	Section 1	Abel Moetji	Abel Moetji	100%	
5	Site 4	Continuous Miner	CM001T	Section 5	Abel Moetji	Abel Moetji	100%	
6	Site 1	Continuous Miner	CM2	Section 2	Lwazi Zondo	Rossouw Snyders	100%	
7	Site 4	Continuous Miner	CM002T	HF2	Quintin Coetzee	Quintin Coetzee	100%	
8	Site 1	Continuous Miner	CM3	Section 3	Chris Bekker	Abel Moetji	100%	
9	Site 4	Continuous Miner	CM003T	HF3	Rossouw Snyders	Rossouw Snyders	100%	
10	Site 1	Continuous Miner	CM4	Section 4	Otto Strydom	Willem Harmse	44%	
11	Site 1	Feeder Breaker	FB1	Section 1	Thabo Mokoena	Quintin Coetzee	100%	
12	Site 2	Feeder Breaker	FB1	Section 1	Document Controller	Abel Moetji	56%	
13	Site 1	Feeder Breaker	FB2	Section 2	Lwazi Zondo	Rossouw Snyders	100%	

Explosion Protected Apparatus Management System Description



Centralized Control:
All certificates are stored in a single, structured system, reducing duplication and loss.

Version Management & Compliance:
Automatically track versions to ensure access to the latest valid certificates and history of previous versions.

Accountability:
Notifications sent to certificate owners and heads of department for missing or outdated certificates. Ownership and compliance status always known.

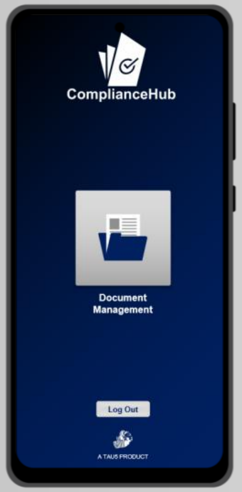
Automated Non-Compliance Reporting:
Flag missing, invalid or expired certificates to maintain compliance.

The Digital Warehouse:
A controlled repository that stores approved digital certificates and allows verified certificate records to be transferred to the specific asset as components are installed or replaced, ensuring traceable and continuous compliance throughout the equipment lifecycle.

User Access Points

Desktop and Web Application.

Android-Based Mobile Application.



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THANK YOU.